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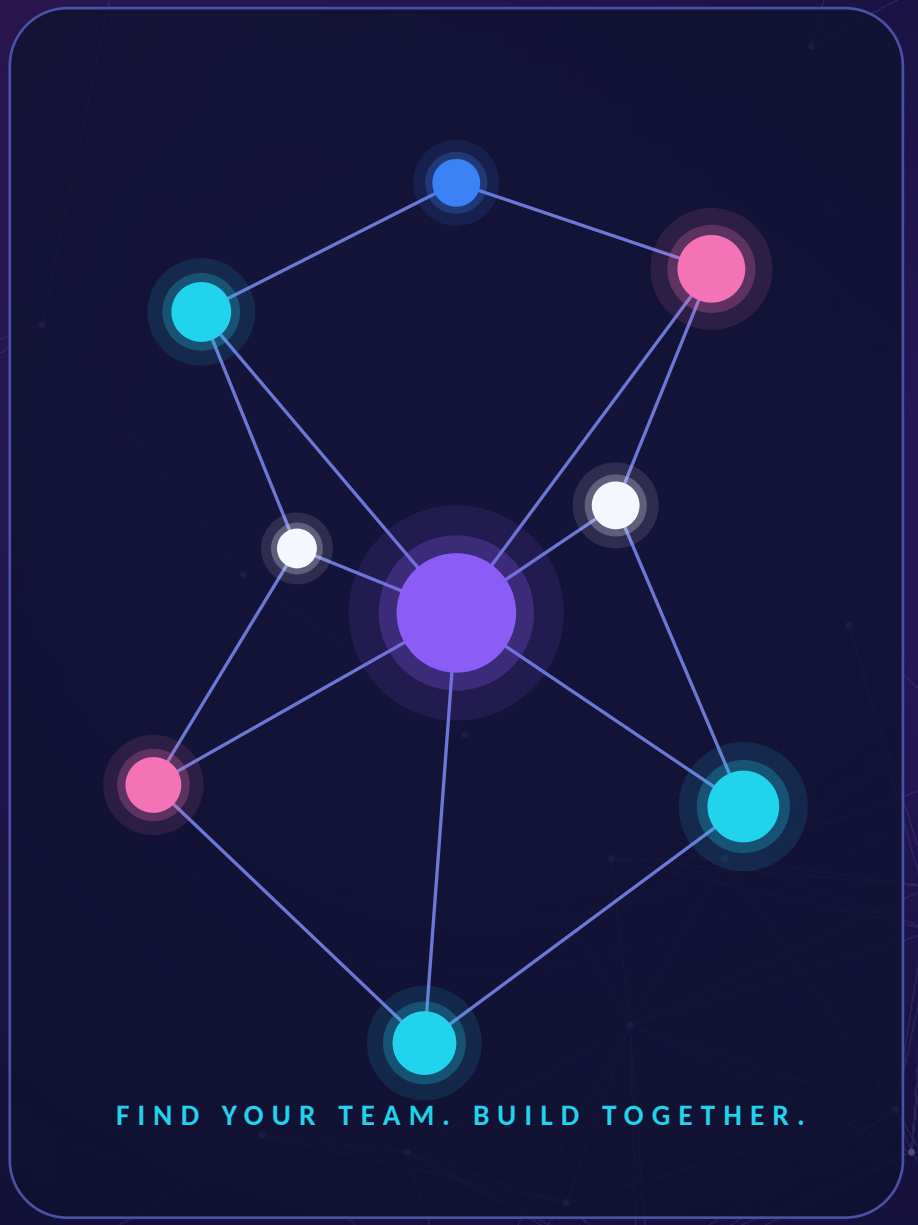
Synapse

A Full-Stack Developer Matchmaking and Hackathon Collaboration Platform

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ABSTRACT

PROBLEM

Finding compatible technical teammates for hackathons and collaborative projects is highly fragmented.

SOLUTION

A centralized, skill-based matchmaking platform designed exclusively for developers.

FUNCTIONALITY

Real-time teammate discovery, hackathon directory, role-based filtering and secure user portfolios.

OUTCOME

Streamlined team formation, higher hackathon participation and balanced team skill sets.

BUILT WITH

N

Next.js 14

Framework + API

P

Prisma

Type-safe ORM

M

MongoDB Atlas

Cloud database

T

Tailwind CSS

Responsive UI

03

OBJECTIVES

1

Full-stack web app

Develop a responsive, full-stack web application for developer collaboration.

2

Secure access

Implement secure user authentication and dynamic portfolio management.

3

Filtering engine

Match users based on specific frameworks and roles.

4

Hackathon dashboard

Integrate a centralized dashboard for upcoming hackathons.

5

Seamless UI/UX

Use modern web standards (React and Tailwind CSS) for a smooth experience.

6

Beyond chat groups

Replace scattered Discord, WhatsApp and LinkedIn searches with one structured platform.

04

INTRODUCTION

BACKGROUND

Hackathons need diverse technical roles to build successful prototypes.

- Frontend
- Backend
- UI/UX
- Data Science

Developers often rely on generic social networks or unorganized chat groups to find teammates.

DIFFICULTIES

- **High friction** verifying tech stacks
- **Poor communication** before events
- **Mismatched goals** within hurriedly formed teams
- **Hard to vouch** for a teammate's real experience

THE NEED

No unified ecosystem validates developer skills and pairs them directly with active hackathons.

Synapse

structured, skill-based team discovery

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LITERATURE SURVEY

EXISTING SYSTEM	APPROACH	KEY FINDING	LIMITATION
Generic Professional Networks	Keyword-based search	Broad reach for hiring	Lacks hackathon-specific team building tools
Dedicated Hackathon Portals	Event registration only	Good event management	Poor peer-to-peer matchmaking features
University Forums	Bulletin board architecture	High local trust	Extremely limited scale and searchability

RESEARCH GAP

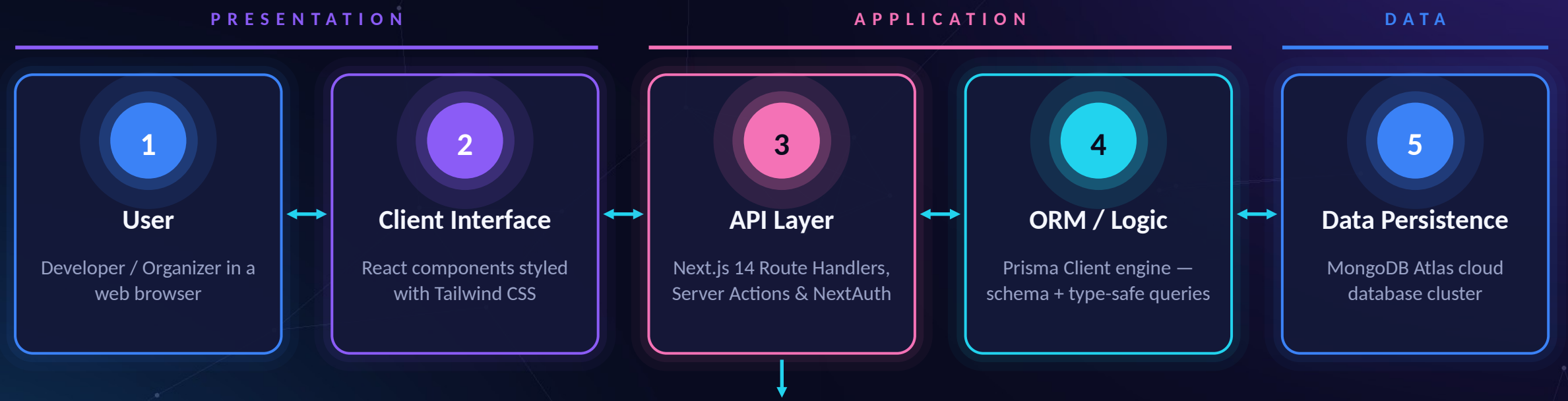
No existing platform combines skill-based filtering with hackathon discovery for developers.

→ **Synapse**

Tech-stack tagging + role filtering + hackathon directory in one platform

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ARCHITECTURE FLOW DIAGRAM



OUTPUT TO THE USER

- ✓ Matched teammate results
- ✓ Hackathon directory listing
- ✓ Updated profile / portfolio
- ✓ Collaboration requests

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PROPOSED SYSTEM

Developer Profiles

tech-stack tags, GitHub links

Matchmaking Engine

queries by tools & roles



Hackathon Directory

active events feed

Secure Auth

NextAuth sessions

MAIN ADVANTAGES

1. **Filter by exact tools**
   Search teammates by Prisma, React, Python and desired roles.
2. **Type-safe queries**
   Prisma ensures fast loading times with strict type safety.
3. **Server-Side Rendering**
   Improves SEO and page performance.
4. **Unified UI**
   Reduces friction in team formation.

08 IMPLEMENTATION DETAILS

DATABASE DESIGN (PRISMA SCHEMA)



Schema sync `npx prisma db push` pushes models to the MongoDB Atlas cluster.

Project layout /app (routes) · /components (UI) · /prisma (schemas)

IMPLEMENTATION STACK

FRONTEND	Next.js 14, React, Tailwind CSS
BACKEND	Next.js API Routes (serverless Node.js)
DATABASE	MongoDB Atlas (NoSQL)
ORM	Prisma Client
AUTH	NextAuth.js

Core logic lives in React Server Components for optimal data fetching.

09

TECHNOLOGY USED

N

FRAMEWORK

Next.js 14 + React

Server-side rendering, routing and API routes in a single codebase.

T

LANGUAGES

TypeScript / JavaScript

Strictly typed code across the stack for reliability.

T

STYLING

Tailwind CSS

Utility-first, responsive, dark-mode-first interface.

M

DATABASE

MongoDB Atlas

Cloud NoSQL cluster with high availability.

P

ORM

Prisma

Schema management and type-safe database queries.

A

AUTHENTICATION

NextAuth.js

Secure onboarding and session persistence.

10 WORKFLOW OF THE PROJECT

1 **User Registration**
Secure sign-up & login
AUTHENTICATION

2 **Tech Stack Setup**
Skills, GitHub, contact
PROFILE ENGINE

3 **Browse Hackathons**
Discover active events
DIRECTORY

4 **Query Teammates**
Filter by tools & roles
MATCHMAKING

5 **Send Request**
Collaboration request

6 **Form Team**
Ready to compete

CORE MODULES

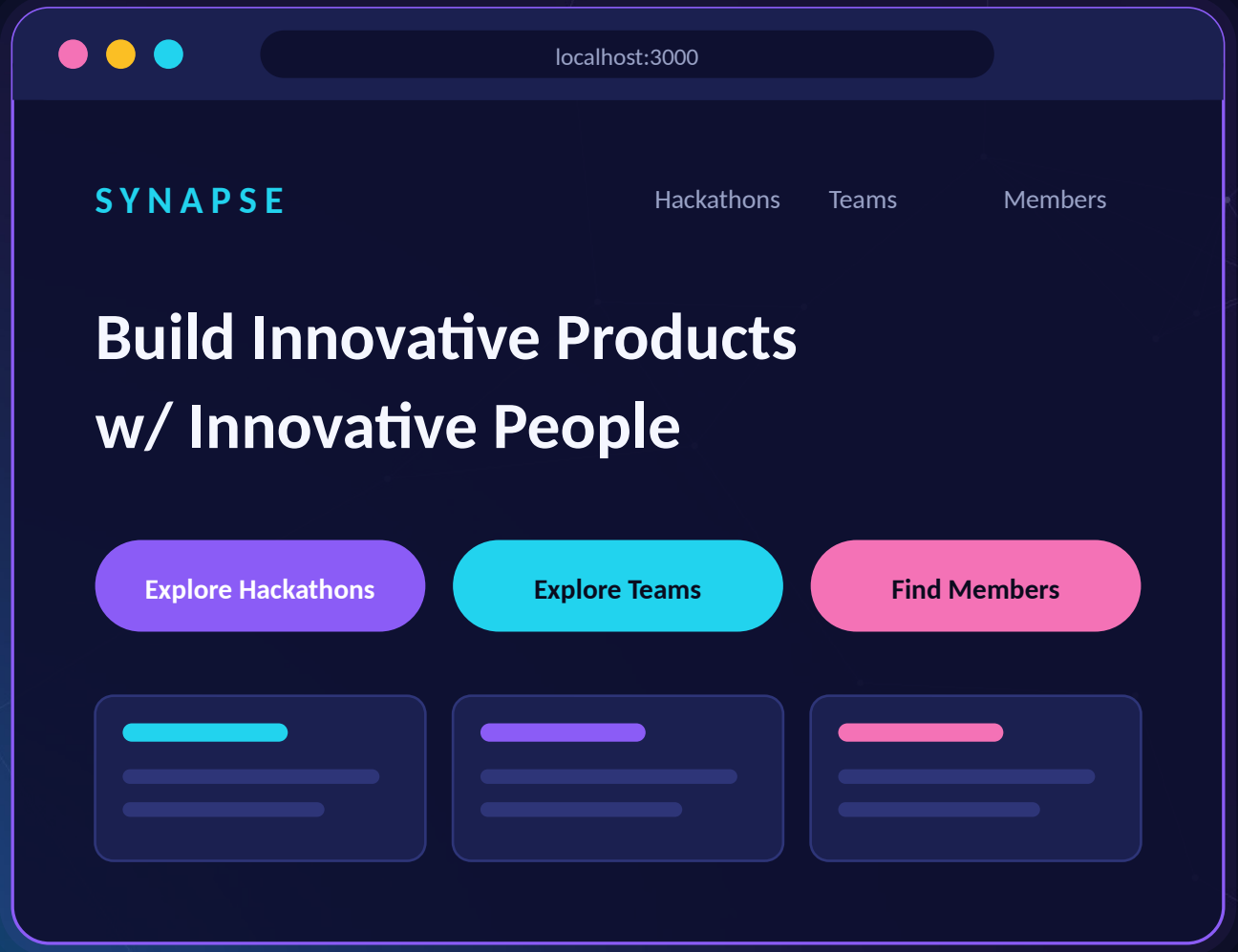
Authentication & Authorization
Secure login, session management and credential verification.

Profile & Portfolio
Tech stack, GitHub links and contact information.

Matchmaking Engine
Queries the database for users matching technical requirements.

Hackathon Directory
Centralized feed of active events for team deployment.

11 RESULTS



UI schematic of the Synapse landing page (dark-mode-first)

Compiles & runs
Unified frontend and backend served on port 3000.

Stable database
Live MongoDB Atlas connection, no fatal exceptions.

Seamless routing
Landing, team exploration and member discovery modules.

Responsive UI
Custom Synapse navigation and hero render correctly.

Current limitations Self-reported skills; collaboration requests are text-based.

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FUTURE ENHANCEMENT

1

TODAY

Core platform live

- Skill-based matchmaking
- Hackathon directory
- Secure portfolios

BUILT

2

NEXT

Closing the gaps

- WebSockets for in-app real-time chat
- GitHub API integration with automated repository analysis

ADDRESSES LIMITATIONS

3

LATER

Smarter matching

- AI-driven project recommendations
- ML-suggested team compositions from historical hackathon-winning patterns

INTELLIGENCE LAYER

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SYNAPSE

Thank You

QUESTIONS & DISCUSSION



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